Design of Enterprise IoT

Tutorial 1 (Solution Scoping)

Read the following use scenario and complete a value proposition document.

IoT Use Case: Smart Toll Collection System

To enhance Taiwan's transportation system and to improve traffic efficiency in all major roads, the government decided to develop an Intelligent System that will be responsible for constructing a smart traffic solution. It is expected that this Intelligent Transport system will collect data from advanced sensors and utilize communications and other technologies to provide real-time information that will increase transportation safety and efficiency.

A major initiative in the context of the above project is to address highway toll collection issues. Previously, the manual toll collection stations were running in all the highways in Taiwan. These toll stations would process more than one million transactions daily. When passing through these toll stations, vehicles would have to reduce their speed to pay the toll with cash or toll tickets. During peak hours and holidays, the slow movement going to these toll stations usually become the source of traffic congestion.

To solve the problem of this toll collection inefficiency, an IoT solution was proposed. The project, called Smart Toll Collection (STC) system, was developed, which used end-to-end technology to provide real-time information on all vehicles travelling on the highways and enhances traffic efficiency. It is considered one of the most important subprojects that have contributed to the development of the overall Intelligent Transport System in Taiwan.

Activity: *Please refer to the 'Value Proposition Template' on Blackboard and discuss the proposed value (and different aspects of needs and risks etc.) for the above Smart Toll System solution.*

You can work on your document by answering the following questions:

- *Substitute*: What people currently do instead?
    - Or, what is the situation without the proposed new solution?
- *Benefits*: What does the solution do?
    - Or, what benefits can it bring to end users (e.g., customers, clients, stakeholders)?
- *Features*: How does the solution work?
- *Experience*: What does it feel like to use the solution (incl. system, product, etc.)?
- *Fears*: What would be the potential risks and/or issues of the proposed solution?
- *Needs*: What are the needs and rational drivers of using this solution?